THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–4: **Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 2310	Highest powers of $2 \times 3 \times 5$, $2 \times 3 \times 7$, $2 \times 3 \times 11$
2	**(b)** 28	$(\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta)$
3	**(a)** -4	$k^2 = 4p^2$; $k = 2p$ gives coincident lines, $k = -2p$ gives parallel lines
4	**(a)** 29	$(\alpha+\beta)^2 - 2\alpha\beta$
5	**(c)** 60	LCM of 1 to 6
6	**(d)** 30	$\alpha\beta(\alpha+\beta)$
7	**(d)** 2	Needs $\frac{k}{4p} = \frac{p}{k} = \frac{k-p}{k}$
8	**(b)** 24	$(\alpha-\beta)^2 = (\alpha+\beta)^2 - 4\alpha\beta$
9	**(d)** 9	HCF of the differences from the remainder
10	**(c)** 7	$\sqrt{(\alpha+\beta)^2 - 4\alpha\beta}$
11	**(c)** 8	Solve for x and y
12	**(d)** 6	$D = 0$ gives $k^2 = 4m$
13	**(a)** Both A and R are true and R is the correct explanation of A.	HCF x LCM = 6 x 36 = 216 = 12 x 18, and R is the general result used to find the LCM.
14	**(a)** Both A and R are true and R is the correct explanation of A.	Each point where the graph meets the x-axis has $y = 0$, i.e. $p(x) = 0$; so R explains A.
15	**(a)** Both A and R are true and R is the correct explanation of A.	$3 + 1 = 4$ and $3 - 1 = 2$; the common point satisfies both equations, which is exactly the solution (R).

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Numbers are $12 \times 3 = 36$ and $12 \times 5 = 60$ (1); LCM $= 180$ (1).	1+1
17	New sum $= \alpha^2 + \beta^2 = 10$ (1); product $= (\alpha\beta)^2 = 9$; polynomial $x^2 - 10x + 9$ (1).	1+1
18	$\frac{k}{8} = \frac{2}{k} \Rightarrow k = \pm 4$ (1); $k = 4$ gives coincident lines, so $k = -4$ (1).	1+1
19	$D = (4)^2 - 4k(1) = 0$ (1); $k = 4$ (1).	1+1
20	Each remainder is short of the divisor by 5 (1); LCM$(12, 18) = 36$, number $= 36 - 5 = 31$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	LCM$(18, 24, 36) = 72$ min (2); rounds 4, 3, 2 (1).	2+1
22	$\alpha + \beta = 3$, $\alpha\beta = 2$ (1); sum $= (\alpha+\beta) + \frac{\alpha+\beta}{\alpha\beta} = \frac{9}{2}$ (1); product $= \alpha\beta + 2 + \frac{1}{\alpha\beta} = \frac{9}{2}$ (1).	1+1+1
23	Put $p = \frac{1}{x}$, $q = \frac{1}{y}$: $2p + 3q = 2$, $3p + 5q = \frac{19}{6}$ (1); eliminating q: $(1)p = \frac{1}{2}$ so $p = \frac{1}{2}$ (1); $q = \frac{1}{3}$ (1); $x = 2$, $y = 3$ (1).	1+1+1+1
24	LHS $= \frac{2(x^2+1)}{x^2-1}$ (1); cross-multiplying: $3(x^2+1) = 5(x^2-1)$ (1); $x^2 = 4$ so $x = \pm 2$ (1).	1+1+1
	OR: Required number $=$ HCF$(1250, 9376)$ (1); HCF$(1250, 9376) = 2$ (2).	1+2
25	$(\alpha+\beta)^2 - (\alpha-\beta)^2 = 4\alpha\beta$ (1); $100 - 16 = 4c$ (1); $c = 21$ (1).	1+1+1
	OR: Need $\frac{a+b}{3} = \frac{a-b}{1} = \frac{14}{7} = 2$ (1); $a + b = 6$, $a - b = 2$ (1); $a = 4$, $b = 2$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) $144 = 2^4 \times 3^2$, $180 = 2^2 \times 3^2 \times 5$ (1). (ii) HCF $= 2^2 \times 3^2 = 36$ chairs per row (1). (iii) Rows: $144/36 + 180/36 = 4 + 5 = 9$ (1). (iv) $216 = 2^3 \times 3^3$; HCF(144, 180, 216) = $2^2 \times 3^2 = 36$ (1); rows $= 4 + 5 + 6 = 15$ (1).	1+1+1+2

27	Let boat x km/h, stream y km/h; put $p=\frac{1}{x-y}$, $q=\frac{1}{x+y}$: $36p+40q=10$, $24p+70q=11$ (2); $p=\frac{1}{6}$, $q=\frac{1}{10}$ (1); $x-y=6$, $x+y=10$ (1); boat 8 km/h, stream 2 km/h (1).	2+1+1+1
	OR: Smaller tap x h, larger $x-5$ h: $\frac{1}{x}+\frac{1}{x-5}=\frac{1}{\frac{10}{3}}$ (2); simplify to a quadratic and solve, rejecting the root less than 5: $x=10$ (2); times 10 h and 5 h (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–4: **Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 2	Substitute the root
2	**(c)** $\frac{\sqrt{12}}{\sqrt{3}}$	$\sqrt{12}/\sqrt{3} = 2$
3	**(a)** 11	$\frac{c}{a}$ for the monic cubic
4	**(c)** 29	$(x+y)^2 - 2xy = 49 - 20$
5	**(a)** $k \le 3$ and $k \neq 0$	$D = 36 - 12k \ge 0$
6	**(b)** $2^3 \times 5^2 \times 3$	Zeros at the end = min(power of 2, power of 5)
7	**(b)** 21	$(\alpha-\beta)^2 = (\alpha+\beta)^2 - 4\alpha\beta$
8	**(b)** 3	Subtract the equations
9	**(b)** $\frac{5}{6}$	$\frac{\alpha+\beta}{\alpha\beta}$
10	**(d)** 144	Highest powers of $2^2 \times 3^2$, $2^4 \times 3$, $2^3 \times 3^2$
11	**(d)** 7	Sum of zeroes $= 9$, so other zero $= 9 - 2$
12	**(a)** -6	$k^2 = 4p^2$; $k = 2p$ gives coincident lines, $k = -2p$ gives parallel lines
13	**(a)** Both A and R are true and R is the correct explanation of A.	$D = 9 - 40 = -31 < 0$, so there are no real roots, as stated in R.
14	**(a)** Both A and R are true and R is the correct explanation of A.	5 is rational and $2\sqrt{3}$ is irrational, so the difference is irrational; R explains it.
15	**(c)** A is true but R is false.	Sum $= -\frac{b}{a} = \frac{8}{2} = 4$, so A is true; R is false because $\frac{c}{a}$ is the product.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Sum $= 6$, product $= k$ (1); $k = 6$ (1).	1+1
17	Adding: $8(x+y) = 88$ so $x+y = 11$ (1); subtracting: $x - y = 3$, so $x = 7$, $y = 4$ (1).	1+1
18	$\alpha+\beta = 6$, $\alpha\beta = 5$ (1); $\alpha^2+\beta^2 = 6^2 - 2(5) = 26$ (1).	1+1
19	If $3\sqrt{7} = r$ were rational, $\sqrt{7} = r/3$ would be rational (1) - a contradiction (1).	1+1
20	Zeroes α and 3α: $4\alpha = 8$ so $\alpha = 2$ (1); $k = 3\alpha^2 = 12$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$D = 25 - 4k \ge 0$ (1); $k \le \frac{25}{4}$ (1); greatest integer $k = 6$ (1).	1+1+1
22	Take 7 common: $7(5 \times 6 + 1)$ (2); $5 \times 6 + 1 = 31 > 1$, so there are two factors greater than 1 (1); composite.	2+1
23	$\sum\alpha = 6$, $\sum\alpha\beta = 11$ (1+1); $\sum\alpha^2 = (\sum\alpha)^2 - 2\sum\alpha\beta = 14$ (1).	1+1+1
24	Intersection $(4, 2)$ (1); x-intercepts 6 and 2 (1); area $= \frac{1}{2} \times 4 \times 2 = 4$ (1).	1+1+1
	OR: Integers x, $x+2$: $x^2 + (x+2)^2 = 74$ (1); $x^2 + 2x - 35 = 0$ (1); $x = 5$; integers 5 and 7 (1).	1+1+1
25	LCM$(20, 28, 35) = 140$ min (2); rounds 7, 5, 4 (1).	2+1
	OR: $\alpha+\beta = 5$, $\alpha\beta = 6$ (1); sum $= (\alpha+\beta) + \frac{\alpha+\beta}{\alpha\beta} = \frac{35}{6}$ (1); product $= \alpha\beta + 2 + \frac{1}{\alpha\beta} = \frac{49}{6}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) $x^2 - 8x + 15 = (x-3)(x-5)$, so zeros are 3 and 5 (1). (ii) Sum $= 8$, product $= 15$ (1). (iii) Twice, at $(3, 0)$ and $(5, 0)$ (1). (iv) Reciprocal zeros $\frac{1}{3}, \frac{1}{5}$: sum $= \frac{8}{15}$, product $= \frac{1}{15}$, polynomial $x^2 - \frac{8}{15}x + \frac{1}{15}$, i.e. $15x^2 - 8x + 1$ (2).	1+1+1+2

27	Let train u, bus v km/h; $p = \frac{1}{u}$, $q = \frac{1}{v}$: $150p + 210q = \frac{41}{4}$, $240p + 120q = 11$ (2); $p = \frac{1}{30}$, $q = \frac{1}{40}$ (2); train 30 km/h, bus 40 km/h (1).	2+2+1
	OR: Boat speed x: $\frac{48}{x-2} + \frac{48}{x+2} = 7$ (1); $48(2x) = 7(x^2 - 4)$ (1); roots of the resulting quadratic: $x = 14$ or negative (1); speed 14 km/h (1).	1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–4: **Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 14	Needs $\frac{k}{4p} = \frac{p}{k} = \frac{k-p}{k}$
2	**(c)** 32	$(\alpha-\beta)^2 = (\alpha+\beta)^2 - 4\alpha\beta$
3	**(c)** 24	HCF of the differences from the remainder
4	**(a)** 56	$\alpha\beta(\alpha+\beta)$
5	**(a)** 7	Solve for x and y
6	**(a)** 8	$D = 0$ gives $k^2 = 4m$
7	**(d)** 2	$18 = 2 \times 3^2$; multiply by the primes that occur to an odd power
8	**(d)** 7	$\sqrt{(\alpha+\beta)^2 - 4\alpha\beta}$
9	**(d)** 6	Intercepts 4 and 3; area $= \frac{1}{2} \times 4 \times 3$
10	**(b)** 2	Substitute the root
11	**(d)** $\frac{\sqrt{75}}{\sqrt{3}}$	$\sqrt{75}/\sqrt{3} = 5$
12	**(c)** $\frac{37}{6}$	$\frac{\alpha^2+\beta^2}{\alpha\beta}$
13	**(c)** A is true but R is false.	$\frac{1}{2} = \frac{2}{4} \neq \frac{4}{12}$, so the lines are parallel and there is no solution: A true; R is false.
14	**(c)** A is true but R is false.	$D = 36 - 36 = 0$ gives equal roots (3 and 3): A true; R is false.
15	**(c)** A is true but R is false.	$\sqrt{2} \times \sqrt{8} = \sqrt{16} = 4$ is rational, so A is true; R is false (this is a counter-example).

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	x-intercept $(4, 0)$, y-intercept $(0, 6)$ (1); area $= \frac{1}{2} \times 4 \times 6 = 12$ (1).	1+1
17	Let number x: $x + \frac{1}{x} = \frac{5}{2}$ (1); $x = 2$ or $x = \frac{1}{2}$ (1).	1+1
18	Numbers are $15 \times 4 = 60$ and $15 \times 7 = 105$ (1); LCM $= 420$ (1).	1+1
19	New sum $= \alpha^2 + \beta^2 = 13$ (1); product $= (\alpha\beta)^2 = 36$; polynomial $x^2 - 13x + 36$ (1).	1+1
20	$\frac{k}{12} = \frac{3}{k} \Rightarrow k = \pm 6$ (1); $k = 6$ gives coincident lines, so $k = -6$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Put $p = \frac{1}{x}$, $q = \frac{1}{y}$: $3p + 4q = \frac{11}{4}$, $5p + 2q = \frac{9}{4}$ (1); eliminating q: $(-14)p = -\frac{7}{2}$ so $p = \frac{1}{4}$ (1); $q = \frac{1}{2}$ (1); $x = 4$, $y = 2$ (1).	1+1+1+1
22	LHS $= \frac{2(x^2+4)}{x^2-4}$ (1); cross-multiplying: $5(x^2+4) = 13(x^2-4)$ (1); $x^2 = 9$ so $x = \pm 3$ (1).	1+1+1
23	Required number $=$ HCF$(1152, 1392)$ (1); HCF$(1152, 1392) = 48$ (2).	1+2
24	$(\alpha+\beta)^2 - (\alpha-\beta)^2 = 4\alpha\beta$ (1); $144 - 36 = 4c$ (1); $c = 27$ (1).	1+1+1
	OR: Need $\frac{a+b}{5} = \frac{a-b}{3} = \frac{12}{4} = 3$ (1); $a + b = 15$, $a - b = 9$ (1); $a = 12$, $b = 3$ (1).	1+1+1
25	$D = 49 - 4k \geq 0$ (1); $k \leq \frac{49}{4}$ (1); greatest integer $k = 12$ (1).	1+1+1
	OR: Take 8 common: $8(6 \times 7 + 1)$ (2); $6 \times 7 + 1 = 43 > 1$, so there are two factors greater than 1 (1); composite.	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) $3n + 2p = 70$ and $2n + 3p = 65$ (1). (ii) Multiply by 3 and 2: $9n + 6p = 210$, $4n + 6p = 130$; subtract: $5n = 80$, so $n =$ ₹ 16 (1). (iii) $3(16) + 2p = 70 \Rightarrow p =$ ₹ 11 (1). (iv) 6 notebooks cost ₹ 96; remaining ₹ 104; $104 \div 11 = 9$ remainder 5, so 9 pens (1) and ₹ 5 left (1).	1+1+1+2

27	Let boat x km/h, stream y km/h; put $p = \frac{1}{x-y}$, $q = \frac{1}{x+y}$: $21p + 33q = 10$, $12p + 55q = 9$ (2); $p = \frac{1}{3}$, $q = \frac{1}{11}$ (1); $x - y = 3$, $x + y = 11$ (1); boat 7 km/h, stream 4 km/h (1).	2+1+1+1
	OR: Smaller tap x h, larger $x - 4$ h: $\frac{1}{x} + \frac{1}{x-4} = \frac{1}{\frac{24}{5}}$ (2); simplify to a quadratic and solve, rejecting the root less than 4: $x = 12$ (2); times 12 h and 8 h (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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